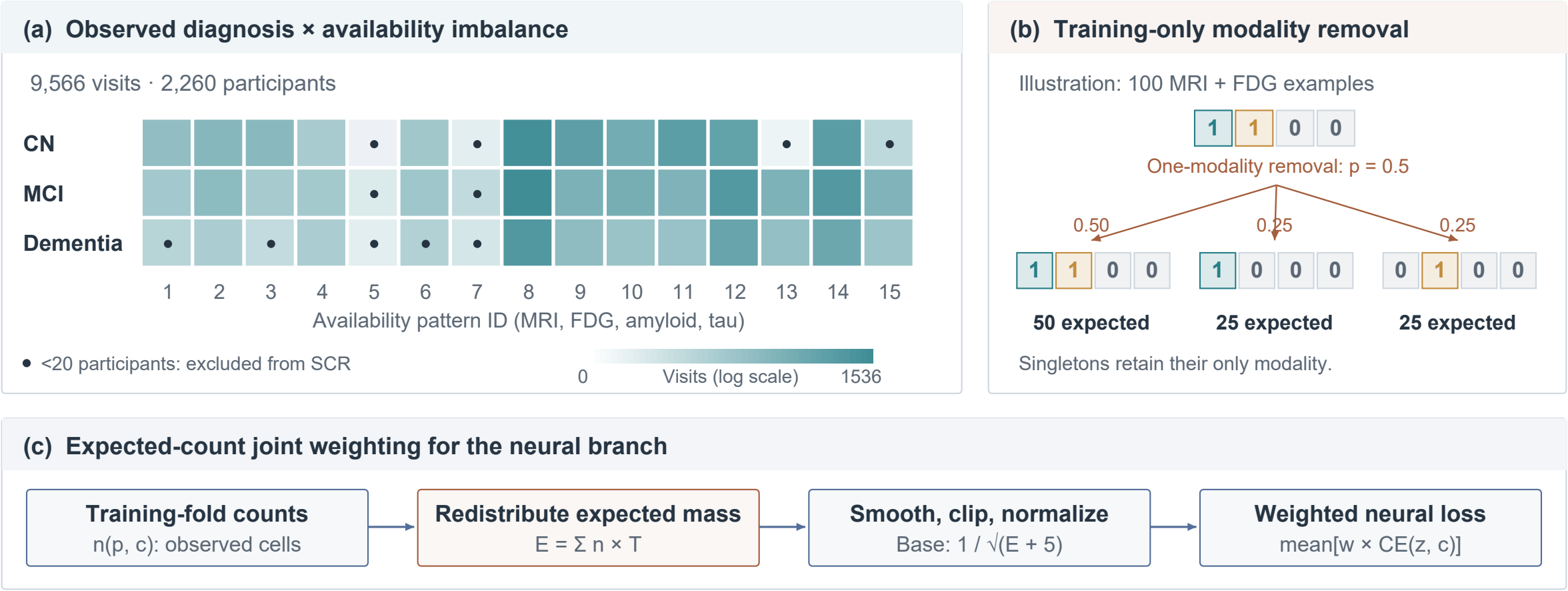

(a) Observed diagnosis × availability imbalance
9,566 visits · 2,260 participants
CN
MCI
Dementia
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Availability pattern ID (MRI, FDG, amyloid, tau)
• <20 participants: excluded from SCR
0
Visits (log scale)
1536
(b) Training-only modality removal
Illustration: 100 MRI + FDG examples
1 1 0 0
One-modality removal: p = 0.5
0.50
0.25
0.25
1 1 0 0
1 0 0 0
0 1 0 0
50 expected
25 expected
25 expected
Singletons retain their only modality.
(c) Expected-count joint weighting for the neural branch
Training-fold counts
n(p, c): observed cells
Redistribute expected mass
E = Σ n × T
Smooth, clip, normalize
Base: 1 / √(E + 5)
Weighted neural loss
mean[w × CE(z, c)]